How BARCS Finds Important Genes

Chapter 1: The ideas

Counts, proportions, noise and the design matrix

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About this guide. This guide explains how the BARCS computer program works. You need only arithmetic, fractions, percents, powers and square roots. The guide teaches logarithms when it uses them. Both chapters use the same small set of numbers, so you can follow every step. The guide uses the writing rules of ASD-STE100 Simplified Technical English.

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The experiment: a CRISPR screen

Each cell in your body has about 20,000 genes. A gene is a set of instructions. Scientists want to know which genes a cell needs, for example to survive a cancer drug.

A **CRISPR screen** tests thousands of genes at the same time. It works in four steps:

1. Scientists make a **library** of short pieces of RNA, called **guides**. Each guide switches off one gene.
2. They put one guide into each cell. Now each cell has one gene switched off.
3. They grow the cells. Some cells grow under normal conditions (**control**). Other cells grow with a drug (**drug**).
4. A sequencing machine reads the guides in each group. The machine counts how many times it reads each guide. This number is a **read count**.

If cells need a gene to survive the drug, the cells that lose that gene die. Their guide becomes rare in the drug group. So the guide count goes down.

Remember

BARCS does not look at cells directly. It looks at read counts. A guide that becomes rare tells us that its gene is important.

Scientists repeat each group more than one time. Each repeat is a **replicate**. Each sample that the machine reads is a **library**. In this guide, we use four libraries: two control replicates and two drug replicates.

The data

The machine gives two numbers for each library:

- the read count of our guide, and
- the total read count of all guides in that library.

Table 1 shows the data for one guide. We use these numbers in all sections of this guide.

What the symbols mean

K	The read count of one guide in one library. Library 1 has $K = 180$.
N	The total read count of all guides in one library. Library 1 has $N = 1{,}000{,}000$.
i	The library number (1, 2, 3 or 4). We write K_i for the count in library i. So $K_2 = 240$.

Table 1: Read counts for one guide in four libraries.

Library	Group	Guide count K	Total count N
1	control	180	1,000,000
2	control	240	1,200,000
3	drug	120	1,000,000
4	drug	80	800,000

Remember

Always record N before you remove any guides from the table. If you remove guides first, the totals become wrong. Then all the later numbers become wrong too.

Step 1: Turn counts into proportions

Library 2 has more reads than library 1 because the machine read it more. So we cannot compare raw counts. We divide each count by its library total. The result is a **proportion**:

$$p = \frac{K}{N}.$$

Proportions are very small numbers. So we multiply by one million. The result is the count **per million** reads.

Worked example 1: proportions

Library 1: $p = 180 \div 1{,}000{,}000 = 0.000180$ = 180 per million
Library 2: $p = 240 \div 1{,}200{,}000 = 0.000200$ = 200 per million
Library 3: $p = 120 \div 1{,}000{,}000 = 0.000120$ = 120 per million
Library 4: $p = 80 \div 800{,}000 = 0.000100$ = 100 per million

The two control libraries give 180 and 200 per million. The two drug libraries give 120 and 100 per million. The guide is less common with the drug.

What the symbols mean

p The proportion of all reads that belong to the guide. It is always between 0 and 1.

Step 2: Compare the groups with a fold change

We want one number that tells how much the guide changes. First, we combine the replicates of each group. We add their counts and add their totals:

$$p_{\text{control}} = \frac{180 + 240}{1{,}000{,}000 + 1{,}200{,}000} = \frac{420}{2{,}200{,}000} = 0.000191 \text{ (191 per million)},$$

$$p_{\text{drug}} = \frac{120 + 80}{1{,}000{,}000 + 800{,}000} = \frac{200}{1{,}800{,}000} = 0.000111 \text{ (111 per million)}.$$

The **fold change** is the drug proportion divided by the control proportion:

$$\text{fold change} = \frac{p_{\text{drug}}}{p_{\text{control}}} = \frac{0.000111}{0.000191} = 0.58.$$

A fold change of 0.58 means that the guide keeps only 58% of its control level. It loses 42%.

Remember

- Fold change $= 1$: no change.
- Fold change < 1: the guide becomes rarer. Cells may need this gene.
- Fold change > 1: the guide becomes more common. Losing this gene may help the cells.

Logarithms make changes symmetric

A fold change of 2 means "double". A fold change of 0.5 means "half". These changes have the same size, but the numbers 2 and 0.5 do not look equal. A **logarithm** fixes this problem.

The **base-2 logarithm**, $\log_2$, tells how many times you double.

Fold change	0.25	0.5	1	2	4
$\log_2$(fold change)	-2	-1	0	1	2

Now "double" is $+1$ and "half" is -1. The sizes look equal.

Worked example 2: log fold change

Our fold change is 0.58. It is between 0.5 and 1. So $\log_2(0.58)$ is between -1 and 0.
A calculator gives $\log_2(0.58) = -0.78$.
The guide drops by 0.78 "halvings".

BARCS uses the **natural logarithm**, ln. It counts powers of the number $e = 2.718\ldots$ instead of powers of 2. The idea is the same. To change back from ln, you raise e to that power: if $\ln(x) = b$, then $x = e^b$. For our guide, $\ln(0.58) = -0.54$.

Step 3: Odds and log-odds

BARCS works with **odds** instead of proportions. The odds compare the reads that belong to the guide with the reads that do not:

$$\text{odds} = \frac{p}{1-p}.$$

The **log-odds** is the natural logarithm of the odds:

$$\text{log-odds} = \ln\left(\frac{p}{1-p}\right).$$

Worked example 3: odds and log-odds

Control: $p = 0.000191$. So odds $= 0.000191 \div 0.999809 = 0.000191$.
The odds and the proportion are almost equal. This is true for all small proportions, because $1 - p$ is almost 1.
Log-odds for control: $\ln(0.000191) = -8.56$.
Log-odds for drug: $\ln(0.000111) = -9.10$.
Difference: $-9.10 - (-8.56) = -0.54$.
The difference of the log-odds is -0.54. This is the same as ln(fold change) from Step 2.

Remember

For guides, a difference in log-odds is almost exactly the natural log of the fold change. So you can read BARCS results as log fold changes.

Why use log-odds at all? A proportion must stay between 0 and 1. A log-odds can be any number, positive or negative. This property lets BARCS add effects together safely, as Step 5 shows.

Step 4: Two kinds of noise

The two control libraries gave 180 and 200 per million. They are not equal. Replicates always differ a little. We call this difference **noise**. BARCS must know how big the noise is. Then it can tell a real change from a chance change.

Guide counts have two kinds of noise:

1. **Counting noise.** The machine reads a random sample of all the guides. Think of a bag with a million marbles. You take out a handful. You do not get exactly the same number of red marbles each time.
2. **Biological noise.** Each replicate is a different group of living cells. The cells grow a little differently each time.

BARCS measures the size of the noise with the **variance**. The variance is the average squared distance of a value from its expected value. The **standard deviation** is the square root of the variance. It is the typical distance in the same units as the counts.

This step builds the variance formula of BARCS in four parts. In all formulas, the sign $\times$ means "multiply". So $N \times \mu$ means "N multiplied by μ".

What the symbols mean

N	The library total, as before. Our example uses $N = 1{,}000{,}000$.
μ	"mu". The proportion of the guide that we expect in the library. For the control group, our best value is $\mu = 0.000191$. It is like p from Step 1, but it is the expected value, not the value that we measure.
ρ	"rho". A number that measures the biological noise. If $\rho = 0$, only counting noise exists. A bigger ρ means more biological noise. Part 3 explains it.

Part 1: The expected count

The machine reads N reads. Each read belongs to our guide with the chance μ. So we expect

$$\text{expected count} = N \times \mu.$$

Worked example 4a: expected count

$N \times \mu = 1{,}000{,}000 \times 0.000191 = 191$ reads.
The real counts are near 191, but not exactly 191. The variance tells how far they normally are from 191.

Part 2: Counting noise

Start with **one read**. It belongs to our guide (value 1) or it does not (value 0). We need the variance of this one value. We do it in three steps: first with five marbles, then the reason for the squares, then the general formula.

Step A: Five marbles. Put five marbles in a bag: one red and four blue. Give the red marble the value 1 and each blue marble the value 0. So the chance of red is $\mu = 1/5 = 0.2$. The average value is $(1 + 0 + 0 + 0 + 0)/5 = 0.2$. This average is μ.

Now find the distance of each marble from the average:

Marble	Value	Distance (value − 0.2)	Squared distance
red	1	+0.8	0.64
blue	0	−0.2	0.04
blue	0	−0.2	0.04
blue	0	−0.2	0.04
blue	0	−0.2	0.04
Sum		$0.8 - 0.8 = 0$	0.80
Average (sum ÷ 5)		0	**0.16**

Look at the distance column. The positive and the negative distances cancel. Their sum is 0. This is always true: the distances from an average always add up to 0. So the plain average distance cannot measure the spread. It is 0 for every bag.

To stop the cancelling, we square each distance. A square is never negative. The average squared distance is $0.80/5 = 0.16$. This is the **variance** of one marble.

Step B: Why squares, and not only the size of the distance? We could also drop the minus signs and use the size of each distance. But BARCS needs a measure that **adds up**: the spread of a sum of independent values must be the sum of their spreads. Part 2 adds up a million reads, and Part 4 adds two kinds of noise. Squares have this property. Sizes do not.

Worked example 4b: two coins

Throw two fair coins. Heads = 1, tails = 0. Each coin has $\mu = 0.5$. The four results are equally likely.

Coins	Total	Distance from 1	Squared distance
tails, tails	0	−1	1
tails, heads	1	0	0
heads, tails	1	0	0
heads, heads	2	+1	1

- **Variance of the total:** $(1 + 0 + 0 + 1)/4 = 0.5$.
- **Variance of one coin:** distances are ± 0.5, squared 0.25. Two coins: $0.25 + 0.25 = 0.5$. **The same.** Squares add up.
- **Size of distance, total:** $(1 + 0 + 0 + 1)/4 = 0.5$.
- **Size of distance, one coin:** 0.5. Two coins: $0.5 + 0.5 = 1$. **Not the same** as 0.5. Sizes do not add up.

The statistician Ronald Fisher gave the name *variance* to the average squared distance in

1918. He chose it because the variances of independent values add up.[1] The idea of squared distances is older. In 1805 Adrien-Marie Legendre fitted lines by making the sum of squared distances as small as possible ("least squares"), and Carl Friedrich Gauss used the same idea.[2] BARCS still uses least squares in Chapter 2.

Step C: The general formula. Now replace "one marble in five" by any chance μ. The table has only two kinds of rows:

Value	How often (chance)	Distance from μ	Squared distance
1	μ	$1-\mu$	$(1-\mu)^2$
0	$1-\mu$	$-\mu$	μ^2

The average squared distance is each squared distance multiplied by how often it happens, then added:

$$\text{variance of one read} = \mu \times (1-\mu)^2 + (1-\mu) \times \mu^2.$$

In the five-marble bag: $0.2 \times 0.8^2 + 0.8 \times 0.2^2 = 0.128 + 0.032 = 0.16$. This is the same 0.16 as in the table. (The table gave $0.64 \times \frac{1}{5} + 0.04 \times \frac{4}{5}$.)

Step D: From the long formula to $\mu \times (1-\mu)$. We now open every bracket, one step at a time. The right column does the same step with $\mu = 0.2$, so you can check each line.

First, open the square. A square means "multiply by itself": $(1-\mu)^2 = (1-\mu) \times (1-\mu)$. Multiply each part of the first bracket by each part of the second bracket:

$$(1-\mu) \times (1-\mu) = 1 \times 1 \;-\; 1 \times \mu \;-\; \mu \times 1 \;+\; \mu \times \mu = 1 - 2 \times \mu + \mu^2.$$

Check: $0.8 \times 0.8 = 0.64$, and $1 - 2 \times 0.2 + 0.04 = 0.64$.

[1] R. A. Fisher (1918). The correlation between relatives on the supposition of Mendelian inheritance. *Transactions of the Royal Society of Edinburgh* 52: 399–433.

[2] S. M. Stigler (1986). *The History of Statistics.* Harvard University Press, chapter 1.

Then, expand everything.

With letters	**With $\mu = 0.2$**
1. $\mu \times (1-\mu)^2 + (1-\mu) \times \mu^2$	$0.2 \times 0.64 + 0.8 \times 0.04$
The formula from Step C.	
2. $\mu \times (1 - 2 \times \mu + \mu^2) + (1-\mu) \times \mu^2$	$0.2 \times 0.64 + 0.8 \times 0.04$
Put in the opened square.	
3. $(\mu - 2 \times \mu^2 + \mu^3) + (\mu^2 - \mu^3)$	$(0.2 - 0.08 + 0.008) + (0.04 - 0.008)$
Multiply μ into the first bracket and μ^2 into the second.	
4. $\mu - 2 \times \mu^2 + \mu^2 + \mu^3 - \mu^3$	$0.2 - 0.08 + 0.04 + 0.008 - 0.008$
Put the same powers next to each other.	
5. $\mu - \mu^2$	$0.2 - 0.04 = 0.16$
$-2 \times \mu^2 + \mu^2 = -\mu^2$, and $+\mu^3 - \mu^3 = 0$.	
6. $\mu \times (1-\mu)$	$0.2 \times 0.8 = 0.16$
Take μ out: $\mu - \mu \times \mu = \mu \times (1-\mu)$.	

Every line gives 0.16. So the long formula and the short formula are always equal:

$$\mu \times (1-\mu)^2 + (1-\mu) \times \mu^2 = \mu \times (1-\mu).$$

A shorter way. Both parts of the long formula contain $\mu \times (1-\mu)$. Take this common part out:

$$\mu \times (1-\mu) \times \Big[(1-\mu) + \mu\Big] = \mu \times (1-\mu) \times 1 = \mu \times (1-\mu).$$

The bracket is 1 because $(1-\mu) + \mu = 1$. Check: $0.2 \times 0.8 \times (0.8 + 0.2) = 0.16$.

Worked example 4c: 100 reads

100 reads: The reads are independent. So their variances add up (Step B). 100 reads with $\mu = 0.2$ have the variance $100 \times 0.16 = 16$. The expected count is $100 \times 0.2 = 20$. The standard deviation is $\sqrt{16} = 4$.
Check by repeating. We took 100 marbles ten times. The red counts were 17, 19, 21, 25, 17, 25, 27, 22, 21 and 14. They are near 20, and they usually differ from 20 by about 4.

For N reads, the variances of the N reads add up. So the counting noise is

$$\text{counting noise} = N \times \mu \times (1-\mu).$$

For our guide: $1{,}000{,}000 \times 0.000191 \times 0.999809 = 191$.

Part 3: Biological noise — the proportion wobbles

Counting noise assumes that every replicate has exactly the same true proportion μ. But each replicate is a different group of living cells. Some cells grow a little faster, and some die by chance. So the true proportion of the guide is a little different in each replicate.

What is the wobble? We write p for the true proportion of the guide in one replicate. (This is the proportion in the cells, before the machine reads anything.) Over many replicates, the average of p is μ. The **wobble** is how far p moves away from μ from one replicate to the next. We measure it with the same rule as every variance: the average squared distance. We call it V:

$$V = \text{average of } (p - \mu)^2.$$

Figure 1 shows four replicates. The true proportion is 0.1 in two of them and 0.3 in the other two. The average is $\mu = 0.2$. Each distance is 0.1, each squared distance is 0.01, so the wobble is $V = 0.01$. If every replicate had $p = 0.2$, all the arrows would have length 0, and the wobble would be $V = 0$.

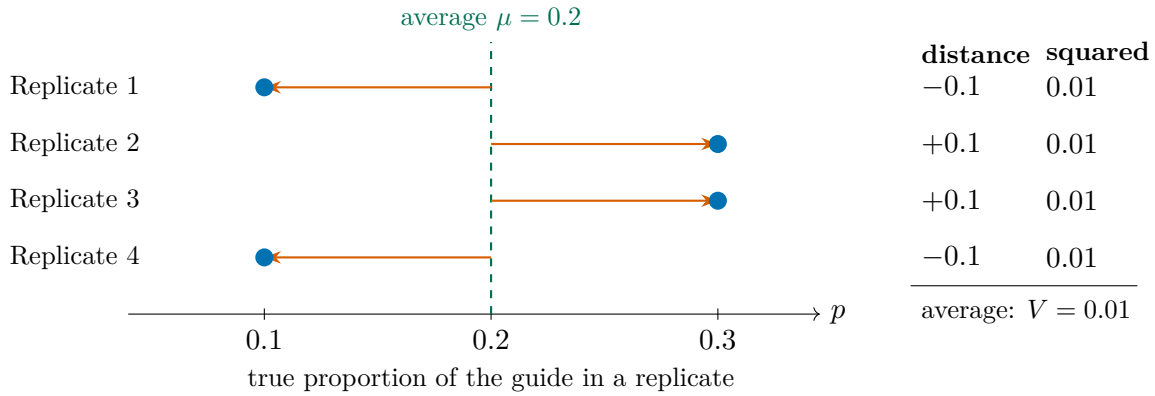


Figure 1: The wobble. Each dot is the true proportion p in one replicate. Each arrow is its distance from the average μ. The wobble V is the average of the squared distances.

Where ρ comes from: two reads from the same replicate. The machine reads many reads from one replicate. Take two of them. Ask this question: *if the first read is our guide, is the second read more likely to be our guide too?*

- **Without a wobble: no.** Every replicate has $p = \mu$. Each read is our guide with the chance μ, whatever the other read is. The chance that both reads are our guide is $\mu \times \mu$.
- **With a wobble: yes.** If the first read is our guide, this replicate probably has a high p. So the second read is also more likely to be our guide. Reads from the same replicate **move together**.

Scientists measure how much two values move together with the **correlation**. A correlation of 0 means that the two values are unrelated. A correlation of 1 means that they are always the same. The number ρ is exactly this: **the correlation between two reads from the same replicate**. That is also why it has the name ρ: it is the usual letter for a correlation.[3]

Now we calculate it. We do it in three steps.

Step 1: the chance that both reads are our guide. First look inside *one* replicate, with the true proportion p. Each read is our guide with the chance p. When p is fixed, the two reads do not affect each other, like two throws of a coin. So the chance that both are our guide is $p \times p$. For example, with $p = 0.3$ it is $0.3 \times 0.3 = 0.09$.

Now look at *all* replicates. Each replicate has its own p. So the chance that both reads are our guide is the average of $p \times p$ over the replicates. In case B, half of the replicates give $0.1 \times 0.1 = 0.01$ and half give $0.3 \times 0.3 = 0.09$. The average is $(0.01 + 0.09)/2 = 0.05$. Without a wobble, it would be $0.2 \times 0.2 = 0.04$. Table 2 shows the three cases of Figure 2.

Case ($\mu = 0.2$)	Both reads are our guide	If there were no wobble ($\mu \times \mu$)	Extra (= covariance)	Correlation ρ = extra $\div$ 0.16
A. $p = 0.2$ always	$0.2 \times 0.2 = 0.04$	0.04	0	0
B. $p = 0.1$ or 0.3	$(0.01 + 0.09)/2 = 0.05$	0.04	0.01	0.0625
C. $p = 0$ or 1	$0.8 \times 0 + 0.2 \times 1 = 0.2$	0.04	0.16	1

Table 2: Two reads from the same replicate. The wobble makes "both reads are our guide" more likely than $\mu \times \mu$. The extra is the covariance. It is always equal to the wobble V.

Step 2: the extra is the covariance, and it equals V. The extra over $\mu \times \mu$ shows how much the two reads move together. Its name is the **covariance**. It is the average of (distance of read 1 from μ) $\times$ (distance of read 2 from μ). Call the two reads X_1 and X_2 (each is 1 or 0). Open the brackets:

$$\begin{aligned}
\text{covariance} &= \text{average of } (X_1 - \mu) \times (X_2 - \mu) \\
&= \text{average of } \Big(X_1 \times X_2 - \mu \times X_2 - X_1 \times \mu + \mu \times \mu\Big) \\
&= \text{average of } X_1 \times X_2 - \mu \times \mu - \mu \times \mu + \mu \times \mu \\
&= (\text{chance that both are our guide}) - \mu \times \mu.
\end{aligned}$$

The third line uses that the average of each read is μ. The last line needs one more fact. The product $X_1 \times X_2$ has only four possible results:

[3] In statistics, ρ is called the *intra-class correlation*: the correlation of two trials in the same group. The beta-binomial model, in which the true proportion wobbles between groups, was described by J. G. Skellam (1948), *Journal of the Royal Statistical Society B* 10: 257–261.

X_1	X_2	$X_1 \times X_2$	
1	1	1	both reads are our guide
1	0	0	
0	1	0	
0	0	0	

So the average of $X_1 \times X_2$ is $1\times$ (chance both are our guide) + $0\times$ (the other chances) = the chance that both are our guide. In case B: $0.05 - 0.04 - 0.04 + 0.04 = 0.05 - 0.04 = 0.01$. Now put in Step 1. The chance that both reads are our guide is the average of $p \times p$, which we write as the average of p^2. So

$$\text{covariance} = \text{average of } p^2 - \mu^2.$$

Case B: $0.05 - 0.04 = 0.01$.

Last, we show that the right side is the wobble V. Start from the definition of V and open the square:

$$\begin{aligned}
V &= \text{average of } (p-\mu) \times (p-\mu) && \text{(definition of } V\text{)} \\
&= \text{average of } \left(p^2 - 2 \times \mu \times p + \mu^2\right) && \text{(multiply out)} \\
&= \text{average of } p^2 - 2 \times \mu \times (\text{average of } p) + \mu^2 && \text{(average each part)} \\
&= \text{average of } p^2 - 2 \times \mu^2 + \mu^2 && \text{(the average of } p \text{ is } \mu\text{)} \\
&= \text{average of } p^2 - \mu^2.
\end{aligned}$$

Case B: $0.05 - 2 \times 0.2 \times 0.2 + 0.04 = 0.05 - 0.08 + 0.04 = 0.01$.

The covariance and V are both equal to the average of $p^2 - \mu^2$. So

$$\text{covariance of two reads} = V.$$

In case B both are 0.01. The wobble of the proportion *is* the covariance of two reads.

Step 3: the correlation. The correlation is the covariance divided by the standard deviations of the two values.[4] Each read has the variance $\mu \times (1-\mu)$ (Part 2), so each standard deviation is $\sqrt{\mu \times (1-\mu)}$. The product of the two standard deviations is $\mu \times (1-\mu)$. So

$$\rho = \frac{\text{covariance}}{\text{SD of read 1} \times \text{SD of read 2}} = \frac{V}{\mu \times (1-\mu)}.$$

Check with case B: $\rho = 0.01 \div 0.16 = 0.0625$. A computer drew 100,000 pairs of reads from case-B replicates and found the correlation 0.06. In case C, the two reads are always the same (both 0 or both 1), and the correlation is 1.

[4] This is Pearson's correlation coefficient: covariance ÷ (standard deviation of value 1 × standard deviation of value 2).

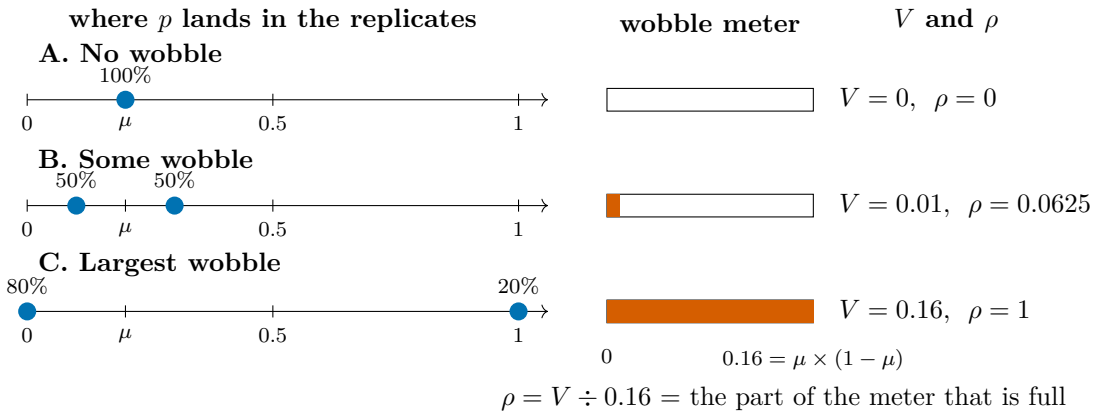


Figure 2: ρ, the correlation of two reads from the same replicate. It is 0 with no wobble and 1 with the largest possible wobble, $\mu \times (1 - \mu)$. In between, ρ tells how full the "meter" is.

Why ρ is never more than 1. A correlation cannot be more than 1. Here this means that the wobble can never be more than $\mu \times (1 - \mu)$. The table shows why, with checks for case B and case C.

Rule	**Case B** ($p = 0.1, 0.3$)	**Case C** ($p = 0, 1$)
a. V = average of $p^2 - \mu^2$	$0.05 - 0.04 = 0.01$	$0.2 - 0.04 = 0.16$
b. $p \times p \leq p$ for every p from 0 to 1	$0.01 \leq 0.1$, $0.09 \leq 0.3$	$0 = 0$, $1 = 1$
c. so: average of $p^2 \leq \mu$	$0.05 \leq 0.2$	$0.2 = 0.2$
d. so: $V \leq \mu - \mu^2 = \mu \times (1 - \mu)$	$0.01 \leq 0.16$	$0.16 = 0.16$

- **Line a.** Open the square: $(p - \mu)^2 = p^2 - 2 \times \mu \times p + \mu^2$. Take the average. The average of p is μ, so the middle part becomes $2 \times \mu \times \mu$. The result is: average of $p^2 - 2 \times \mu^2 + \mu^2$ = average of $p^2 - \mu^2$.
- **Line b.** A number from 0 to 1 does not grow when you multiply it by itself: $0.3 \times 0.3 = 0.09$. It stays the same only for 0 and 1.
- **Line c.** Each p^2 is at most p. So the average of p^2 is at most the average of p, which is μ.
- **Line d.** Put line c into line a.

Turn the formula around. Multiply both sides of the formula for ρ by $\mu \times (1 - \mu)$:

$$\rho = \frac{V}{\mu \times (1 - \mu)} \quad \Longrightarrow \quad \text{covariance of two reads} = V = \rho \times \mu \times (1 - \mu).$$

Check with case B: $0.0625 \times 0.16 = 0.01$. We use this in Part 4.

Worked example 4d: the wobble of our guide

Variance of one read: $\mu \times (1 - \mu) = 0.000191 \times 0.999809 = 0.000191$.
Correlation of two reads: BARCS estimates $\rho = 0.000000386$ (Step 7 in Chapter 2 shows how). Two reads from the same library are almost unrelated.
The wobble (= covariance): $V = \rho \times \mu \times (1 - \mu) = 0.000000386 \times 0.000191 = 0.0000000000737$.
In words: the square root of V is 0.0000086, which is 8.6 per million. So the true proportion of our guide moves by about 8.6 per million between replicates, for example from 191 to 183 or 199 per million.

Part 4: Add up the reads — where $N \times (N - 1)$ comes from

The count K is the number of reads that are our guide. So K is a sum of N values, one for each read: $K = X_1 + X_2 + \cdots + X_N$, where each X is 1 (our guide) or 0 (not our guide).

Two reads first. Part 2 (Step B) said that the variances of *independent* values add up. But reads from the same replicate are not independent: they move together. Look at the distance of the sum from its average. It is the sum of the two distances, $d_1 + d_2$. Square it:

$$(d_1 + d_2)^2 = d_1 \times d_1 + d_2 \times d_2 + 2 \times d_1 \times d_2.$$

Take the average of each part. The average of $d_1 \times d_1$ is the variance of read 1. The average of $d_1 \times d_2$ is the covariance (Part 3, Step 2). So

$$\text{variance of } (X_1 + X_2) = \text{variance} + \text{variance} + 2 \times \text{covariance}.$$

For independent reads the covariance is 0, and we get Step B again.

Worked example 4e: two reads from a case-B replicate

Variance of one read: $0.2 \times 0.8 = 0.16$. Covariance: $V = 0.01$.
Variance of the sum: $0.16 + 0.16 + 2 \times 0.01 = 0.34$.
A computer drew 100,000 pairs and found 0.340.

N reads. Now square the sum of N distances, $(d_1 + d_2 + \cdots + d_N)^2$. Multiply every part by every part. You get two kinds of products:

- **Squares** $d_1 \times d_1$, $d_2 \times d_2$, ...: there are N of them. Each has the average $\mu \times (1 - \mu)$.
- **Pairs** $d_1 \times d_2$, $d_2 \times d_1$, ...: each read pairs with each of the $N - 1$ other reads. So there are $N \times (N - 1)$ of them. Each has the average covariance V.

Example with $N = 3$ reads: the pairs are 1–2, 1–3, 2–1, 2–3, 3–1, 3–2. That is $6 = 3 \times 2$ pairs. (With two reads, there are $2 \times 1 = 2$ pairs. That is the "2×" above.)

So

$$\text{variance} = \underbrace{N \times \mu \times (1-\mu)}_{N \text{ squares}} + \underbrace{N \times (N-1) \times V}_{N\times(N-1) \text{ pairs}}.$$

This is where $N \times (N-1)$ comes from: it is the number of pairs of reads. Every pair of reads from the same replicate adds its covariance V.

The final form. Put in $V = \rho \times \mu \times (1-\mu)$ from Part 3:

$$\text{variance} = N \times \mu \times (1-\mu) + N \times (N-1) \times \rho \times \mu \times (1-\mu).$$

Both parts contain $N \times \mu \times (1-\mu)$. Take it out, as in $a + a \times b = a \times (1+b)$:

$$\boxed{\text{variance} = \underbrace{N \times \mu \times (1-\mu)}_{\text{counting noise}} \times \underbrace{\big(1 + (N-1) \times \rho\big)}_{\text{extra factor from biological noise}}}$$

Figure 3 shows the whole path.

Step	**Formula** / *check with case B, $N = 10$*
1. The wobble	V = average of $(p-\mu)^2$
	$V = 0.01$
2. Covariance of two reads	covariance = (both are our guide) $-\mu \times \mu = V$
	$0.05 - 0.04 = 0.01$
3. Correlation of two reads	ρ = covariance $\div$ (SD $\times$ SD) $= V \div (\mu \times (1-\mu))$
	$0.01 \div 0.16 = 0.0625$
4. Turn it around	$V = \rho \times \mu \times (1-\mu)$
	$0.0625 \times 0.16 = 0.01$
5. Add up N reads	$N \times \mu \times (1-\mu) + N \times (N-1) \times V$
	$1.6 + 90 \times 0.01 = 2.5$
6. Replace V by step 4	$N \times \mu \times (1-\mu) + N \times (N-1) \times \rho \times \mu \times (1-\mu)$
	$1.6 + 90 \times 0.0625 \times 0.16 = 2.5$
7. Take out the common part	$N \times \mu \times (1-\mu) \times (1 + (N-1) \times \rho)$
	$1.6 \times (1 + 9 \times 0.0625) = 1.6 \times 1.5625 = 2.5$

Figure 3: How ρ gets into the variance formula. Steps 1–4 come from Part 3, and step 5 from Part 4. In step 6, the orange V is replaced by $\rho \times \mu \times (1-\mu)$ from step 4. The lines in italics check each step with case B and $N = 10$ reads, where $N \times \mu \times (1-\mu) = 10 \times 0.2 \times 0.8 = 1.6$ and $N \times (N-1) = 90$.

Worked example 4f: check with ten reads

Case B ($\mu = 0.2$, $V = 0.01$, $\rho = 0.0625$) with $N = 10$ reads.

- Squares: $10 \times 0.16 = 1.6$.
- Pairs: $10 \times 9 = 90$ pairs, each 0.01: $90 \times 0.01 = 0.9$.
- Total: $1.6 + 0.9 = 2.5$.
- Boxed formula: $1.6 \times (1 + 9 \times 0.0625) = 1.6 \times 1.5625 = 2.5$.

A computer that drew 100,000 such replicates found the variance 2.498. With $N \times N$ in place of $N \times (N - 1)$, the formula would give 2.6, which is wrong: a read does not form a pair with itself.

Worked example 4g: the total variance of our guide

- Squares: $1{,}000{,}000 \times 0.000191 \times 0.999809 = 191.0$.
- Pairs: $1{,}000{,}000 \times 999{,}999 \times 0.0000000000737 = 73.7$.
- Total: $191.0 + 73.7 = 264.7$.
- Boxed formula: $191 \times (1 + 999{,}999 \times 0.000000386) = 191 \times 1.386 = 264.7$.

Each pair adds very little, but there are almost a million million pairs.
Standard deviation: $\sqrt{264.7} = 16.3$ reads. Counting noise alone gives $\sqrt{191} = 13.8$ reads. Biological noise adds 39% to the variance.

Remember

Look at the extra factor $1 + (N - 1) \times \rho$. It contains N. When the machine reads more, N grows, and the factor grows too. So more reading does not remove biological noise. Only more replicates help with biological noise.

Step 5: The design matrix

A **model** is a rule that predicts a value. BARCS predicts the log-odds of a guide in each library. The model needs to know which group each library belongs to. BARCS gets this information from the **design matrix**.

What a design matrix is

A design matrix is a table of numbers. It has:

- one **row** for each library, and

- one **column** for each β value in the model.

We write the design matrix with the letter X.

For our four libraries, the model has two β values. So X has four rows and two columns:

Library	Group	Column 1: "intercept"	Column 2: "drug"
1	control	1	0
2	control	1	0
3	drug	1	1
4	drug	1	1

What the symbols mean

X	The design matrix. One row for each library, one column for each β.
Intercept	A column of 1s. It belongs to β_0, the starting level. Every library gets the starting level.
Drug	A column that is 1 for a drug library and 0 for a control library. It belongs to β_1, the effect of the drug.
β_0	"beta zero". The log-odds of the guide in the control group.
β_1	"beta one". The change in log-odds from control to drug. This is the **effect**. It is the most important number.

From the design matrix to a prediction

To get the prediction for one library, take the row of that library. Multiply each number in the row by its β. Then add the products:

$$\text{predicted log-odds} = (\text{column 1}) \times \beta_0 + (\text{column 2}) \times \beta_1.$$

This calculation is a **matrix multiplication**. You do it one row at a time. We call the predicted log-odds η ("eta").

Worked example 5: predictions from the design matrix

BARCS finds $\beta_0 = -8.565$ and $\beta_1 = -0.542$ (Step 6 in Chapter 2 shows how).

Library 1 (row 1, 0): $1 \times (-8.565) + 0 \times (-0.542) = -8.565$

Library 2 (row 1, 0): $1 \times (-8.565) + 0 \times (-0.542) = -8.565$

Library 3 (row 1, 1): $1 \times (-8.565) + 1 \times (-0.542) = -9.107$

Library 4 (row 1, 1): $1 \times (-8.565) + 1 \times (-0.542) = -9.107$

The control libraries get β_0 only. The drug libraries get $\beta_0 + \beta_1$. So β_1 is the difference between the groups.

From log-odds back to counts

A prediction in log-odds is hard to read. So we change it back into a proportion, and then into a count. The rule is:

$$\mu = \frac{e^{\eta}}{1 + e^{\eta}}, \qquad \text{expected count} = N \times \mu.$$

This rule undoes the log-odds of Step 3.

Worked example 6: expected counts

Control libraries ($\eta = -8.565$): $e^{-8.565} = 0.000190676$. Then $\mu = 0.000190676 \div 1.000190676 = 0.000190640$, which is 190.6 per million.
Drug libraries ($\eta = -9.107$): $e^{-9.107} = 0.000110840$. Then $\mu = 0.000110827$, which is 110.8 per million.

Library	N	μ (per million)	Expected count $N \times \mu$	Real count K
1	1,000,000	190.6	190.6	180
2	1,200,000	190.6	228.8	240
3	1,000,000	110.8	110.8	120
4	800,000	110.8	88.7	80

The expected counts are near the real counts. The differences are -10.6, $+11.2$, $+9.2$ and -8.7 reads. These differences are the noise.

The reference group

The control group has 0 in the drug column. So control is the **reference group**. The effect β_1 always compares a group with the reference group. If you make drug the reference group, β_1 changes sign. It becomes $+0.542$. The meaning stays the same.

Paired replicates: add a column

Sometimes the replicates are pairs. For example, replicate 1 of control and replicate 1 of drug come from the same batch of cells. Then BARCS can remove the batch difference. You add a column for the batch:

Library	Group	Replicate	Intercept	Replicate 2	Drug
1	control	1	1	0	0
2	control	2	1	1	0
3	drug	1	1	0	1
4	drug	2	1	1	1

Now the model has three β values: the starting level, the batch difference and the drug effect. Each new column costs one degree of freedom (Step 8 in Chapter 2). With four libraries and three columns, only one degree of freedom stays.

Worked example 7: the paired model for our guide

BARCS gives the drug effect $\beta = -0.544$ with standard error 0.143. The effect is almost the same as before (-0.542). But the degrees of freedom drop from 2 to 1. The p-value becomes 0.164 instead of 0.033.
Our two replicates have almost no batch difference ($\beta = 0.005$). So the extra column costs information and gives nothing back. Add a batch column only when the replicates really are pairs.

Numbers instead of groups

A column can also hold a number, for example the day of the experiment. Suppose that you collect cells on day 0, day 7 and day 14, two libraries each. You put the day divided by 14 into a column:

Library	Day	Intercept	Day ÷ 14
1, 2	0	1	0
3, 4	7	1	0.5
5, 6	14	1	1

Now β_1 is the change in log-odds over 14 days. The model draws a straight line through the three days.

Worked example 8: a time course

Suppose that $\beta_1 = -0.70$.
After 14 days: change $= 1 \times (-0.70) = -0.70$. Fold change $= e^{-0.70} = 0.50$. The guide drops to half.
After 7 days: change $= 0.5 \times (-0.70) = -0.35$. Fold change $= e^{-0.35} = 0.70$. The guide drops to 70%.
The model has 6 libraries and 2 columns. So it has $6 - 2 = 4$ degrees of freedom.

Two factors and an interaction

Some experiments change two things. For example, cells can be normal or can have a second gene switched off (“knockout”). And cells can get no drug or the drug. This experiment has four groups. The design matrix has four columns:

Group	Intercept	Knockout	Drug	Knockout × Drug
normal, no drug	1	0	0	0
knockout, no drug	1	1	0	0
normal, drug	1	0	1	0
knockout, drug	1	1	1	1

The last column is the product of the knockout column and the drug column. It is 1 only when both are 1. Its β is the **interaction**. The interaction measures the extra change that happens only when both changes happen together.

Worked example 9: reading an interaction

Suppose that the four β values are -8.5, -0.2, $+0.1$ and -0.7. Multiply each row by the β values and add:

normal, no drug: -8.5

knockout, no drug: $-8.5 + (-0.2) = -8.7$

normal, drug: $-8.5 + 0.1 = -8.4$

knockout, drug: $-8.5 + (-0.2) + 0.1 + (-0.7) = -9.3$

The knockout alone changes the log-odds by -0.2. The drug alone changes it by $+0.1$. If the two effects only added up, the last group would be $-8.5 - 0.2 + 0.1 = -8.6$. The real value is -9.3. The extra -0.7 is the interaction. The guide drops strongly only when both changes happen.

Comparing two treatments: a contrast

Suppose an experiment has three groups: control, drug A and drug B. The design matrix has the columns intercept, drug A and drug B. Then β_A compares drug A with control, and β_B compares drug B with control. To compare drug A with drug B, you subtract: $\beta_A - \beta_B$. This calculation is a **contrast**. In BARCS, the function `bb_contrast()` calculates it and its test.

For example, if $\beta_A = -0.9$ and $\beta_B = -0.4$, then $\beta_A - \beta_B = -0.5$. Drug A lowers the guide more than drug B.

Rules for a good design matrix

1. Make one row for each library, in the same order as the count table.
2. Make one column for each β that you want.
3. Keep more rows than columns. The difference is the degrees of freedom. Two or more is better than one.
4. Do not let two columns carry the same information. For example, suppose that all drug libraries come from batch 2. Suppose also that all control libraries come from batch 1.

Then the batch column and the drug column are equal. BARCS cannot separate the drug effect from the batch effect.

The design matrix in R

In R, you describe the design matrix with a **formula**. The formula `~ group` means "an intercept and a column for each group except the reference group".

The function `model.matrix()` builds X from the formula:

```
> data <- data.frame(group = c("control", "control", "drug", "drug"))
> model.matrix(~ group, data)
  (Intercept) groupdrug
1           1         0
2           1         0
3           1         1
4           1         1
```

Then you give the counts, the totals, the formula and the column name to BARCS:

```
fit <- bb_screen(counts, data, formula = ~ group,
                 term = "groupdrug", totals = totals)
```

For the paired design, use the formula `~ replicate + group`. For the interaction, use `~ knockout * drug`.

Chapter 1 summary

Chapter 1 showed what BARCS measures and how it describes an experiment:

1. It turns read counts into proportions (Step 1).
2. It compares groups with a fold change and its logarithm (Step 2).
3. It works on the log-odds scale (Step 3).
4. It separates counting noise from biological noise (Step 4).
5. It describes the experiment with a design matrix (Step 5).

Chapter 2 shows how BARCS calculates the effect and decides if it is real.

The symbols of Chapter 1

Symbol	Meaning	Value in our example
K	read count of the guide in one library	180, 240, 120, 80
N	total read count of one library	1.0, 1.2, 1.0, 0.8 million
p	proportion, K/N	191 and 111 per million
fold change	$p_{\text{drug}} \div p_{\text{control}}$	0.58
$\log_2$	number of doublings	$\log_2(0.58) = -0.78$
ln	natural logarithm (powers of $e = 2.718$)	$\ln(0.58) = -0.54$
log-odds	$\ln(p/(1-p))$	-8.56 (control)
μ	expected proportion	0.000191 (control)
ρ	biological noise	0.000000386
X	design matrix: a row per library, a column per β	4 rows, 2 columns
β_0	control log-odds in the model	-8.565
β_1	effect of the drug	-0.542
η	predicted log-odds of a library (row of X times β)	-8.565, -9.107

Quiz for Chapter 1

Choose the best answer. The answers and calculations are on the next pages.

Q1. What does one guide do in a CRISPR screen?

(A) It counts the cells.

(B) It adds a drug to the cells.

(C) It reads the DNA.

(D) It switches off one gene.

Q2. A library has $K = 300$ and $N = 1{,}500{,}000$. What is the count per million?

(A) 200

(B) 300

(C) 450

(D) 500

Q3. Why do we divide each count by its library total N?

(A) To make the numbers bigger.

(B) Because the machine reads some libraries more than others.

(C) To remove the biological noise.

(D) Because BARCS cannot use whole numbers.

Q4. A guide has 400 per million in control and 100 per million with a drug. What is the fold change?

(A) 4

(B) 0.4

(C) 0.25

(D) 300

Q5. What is $\log_2(8)$?

(A) 2

(B) 3

(C) 4

(D) 8

Q6. A gene has $\log_2(\text{fold change}) = -1$. What happened to its guide?

(A) It doubled.

(B) It did not change.

(C) It dropped to half.

(D) It dropped to zero.

Q7. For a very small proportion p, which statement is true?

(A) The odds are almost equal to p.

(B) The odds are almost equal to 1.

(C) The odds are twice p.

(D) The odds are always 0.

Q8. In the variance formula, what does $\rho = 0$ mean?

(A) There is no noise at all.

(B) There is only counting noise.

(C) There is only biological noise.

(D) The guide has zero reads.

Q9. Use $N \times \mu \times (1 - \mu)$ with $N = 100$ and $\mu = 0.5$. What is the counting variance?

(A) 25

(B) 50

(C) 0.25

(D) 100

Q10. The machine reads each library ten times more deeply. What happens to the biological noise?

(A) It disappears.

(B) It becomes ten times smaller.

(C) It does not go away. Only more replicates help.

(D) It becomes zero for control guides.

Q11. In the design matrix of our example, which β belongs to the column of 1s?

(A) β_1

(B) the interaction

(C) β_0

(D) ρ

Q12. BARCS gives $\beta_1 = 0.69$ for a guide. Use $e^{0.69} = 2.0$. What does this mean?

(A) The guide halves with the drug.

(B) The guide does not change.

(C) The guide disappears.

(D) The guide doubles with the drug.

Q13. An experiment has 3 control libraries and 3 drug libraries. The formula is `~ group`. What is the size of the design matrix?

(A) 3 rows and 2 columns

(B) 6 rows and 2 columns

(C) 2 rows and 6 columns

(D) 6 rows and 3 columns

Q14. A library has the row $(1, 1)$ in the design matrix. The values are $\beta_0 = -8.0$ and $\beta_1 = -0.5$. What is the predicted log-odds?

(A) -8.5

(B) -8.0

(C) -7.5

(D) 4.0

Q15. In the knockout-and-drug design, which group has a 1 in the interaction column?

(A) normal, no drug

(B) knockout, no drug

(C) normal, drug

(D) knockout, drug

Q16. You add a paired-replicate column to the model of our four libraries. How many degrees of freedom stay?

(A) 3

(B) 2

(C) 4

(D) 1

Answers for Chapter 1

Q1. D. Each guide switches off one gene (Section 1).

Q2. A. $300 \div 1{,}500{,}000 = 0.0002$. Then $0.0002 \times 1{,}000{,}000 = 200$ per million.

Q3. B. Libraries get different numbers of reads. Division by N makes them comparable (Step 1).

Q4. C. $100 \div 400 = 0.25$. The guide keeps one quarter of its control level.

Q5. B. $2 \times 2 \times 2 = 8$. You double three times, so $\log_2(8) = 3$.

Q6. C. -1 means one halving (Step 2).

Q7. A. $1 - p$ is almost 1, so $p/(1 - p)$ is almost p (Step 3).

Q8. B. The extra factor becomes $1 + (N - 1) \times 0 = 1$. Only counting noise stays (Step 4).

Q9. A. $100 \times 0.5 \times (1 - 0.5) = 100 \times 0.5 \times 0.5 = 25$.

Q10. C. The factor $1 + (N - 1) \times \rho$ grows with N. Deeper reading does not remove biological noise (Step 4).

Q11. C. The column of 1s is the intercept. It belongs to β_0, the starting level (Step 5).

Q12. D. The fold change is $e^{0.69} = 2.0$. The guide doubles.

Q13. B. One row for each library ($3 + 3 = 6$). One column for the intercept and one for drug (Step 5).

Q14. A. $1 \times (-8.0) + 1 \times (-0.5) = -8.5$ (Step 5).

Q15. D. The interaction column is knockout times drug. Only $1 \times 1 = 1$ (Step 5).

Q16. D. The model now has 3 columns. So df $= 4 - 3 = 1$ (Step 5).